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FORM TP 2011237

MAY/JUNE 2011

CARIBBEAN EXAMINATIONS COUNCIL

ADVANCED PROFICIENCY EXAMINATION

PHYSICS

UNIT 1 – Paper 02

2 hours 30 minutes

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This paper consists of **SIX** questions, divided into Sections A and B.

2. Section A consists of **THREE** questions. Candidates must attempt **ALL** questions in this section. Answers for this section must be written in the spaces provided in this question paper.

3. Section B consists of **THREE** questions. Candidates must attempt **ALL** questions in this section. Answers for this section must be written in the spaces provided in this question paper.

4. Diagrams may be drawn in pencil but **all writing** must be in **ink**.

5. All working **MUST** be **CLEARLY** shown.

6. The use of silent non-programmable calculators is permitted, but candidates should note that the use of an inappropriate number of figures in answers will be penalised.

LIST OF PHYSICAL CONSTANTS

Universal gravitational constant	G	=	6.67×10^{-11} N m^2 kg^{-2}
Acceleration due to gravity	g	=	9.80 m s^{-2}
1 Atmosphere	Atm	=	1.00×10^5 N m^{-2}
Boltzmann's constant	k	=	1.38×10^{-23} J K^{-1}
Density of water		=	1.00×10^3 kg m^{-3}
Specific heat capacity of water		=	4200 J kg^{-1} K^{-1}
Specific latent heat of fusion of ice		=	3.34×10^5 J kg^{-1}
Specific latent heat of vaporization of water		=	2.26×10^6 J kg^{-1}
Avogadro's constant	N_A	=	6.02×10^{23} per mole
Molar gas constant	R	=	8.31 J K^{-1} mol^{-1}
Stefan-Boltzmann's constant	σ	=	5.67×10^{-8} W m^{-2} K^{-4}
Speed of light in free space	c	=	3.00×10^8 m s^{-1}

SECTION A

Answer ALL questions.

You MUST write your answers in the spaces provided in this answer booklet.

1. (a) Figure 1 shows a skydiver of mass m falling under the influence of gravity and a drag force F_d that is proportional to v^n ($F_d = bv^n$), where v is the velocity of the skydiver as she falls, b is a constant and n is an integer.

Figure 1

(i) Next to Figure 1 draw a free body diagram to show the forces acting on the skydiver. **[1 mark]**

(ii) Hence show that when her downward acceleration is a, the equation of motion for the skydiver can be written as

$$g - a = \frac{b}{m} \quad v^n .$$

[2 marks]

(b) Table 1 shows data recorded for the acceleration and velocity of the skydiver as she underwent free fall.

TABLE 1

Acceleration a/m s^{-2}	**Velocity** v/ms^{-1}	$g - a$/m s^{-2}	lg $(g - a)$	lg v
9.41	10			
8.91	15			
8.24	20			
7.36	25			
6.28	30			
5.00	35			

(i) Complete Table 1 by filling in the blank columns. **[3 marks]**

(ii) On page 4, plot a graph of lg $(g - a)$ vs lg v. **[3 marks]**

(iii) Calculate the gradient of the graph and hence determine a value for n (to the nearest integer).

[3 marks]

(iv) Calculate the terminal velocity of a skydiver with a mass of 78.5 kg, given that $b = 0.251$ kg m^{-1}.

[3 marks]
Total 15 marks

NOTHING HAS BEEN OMITTED.

2. (a) Figure 2 shows a loudspeaker sounding a constant note placed above a long vertical tube containing water which slowly runs out at the lower end.

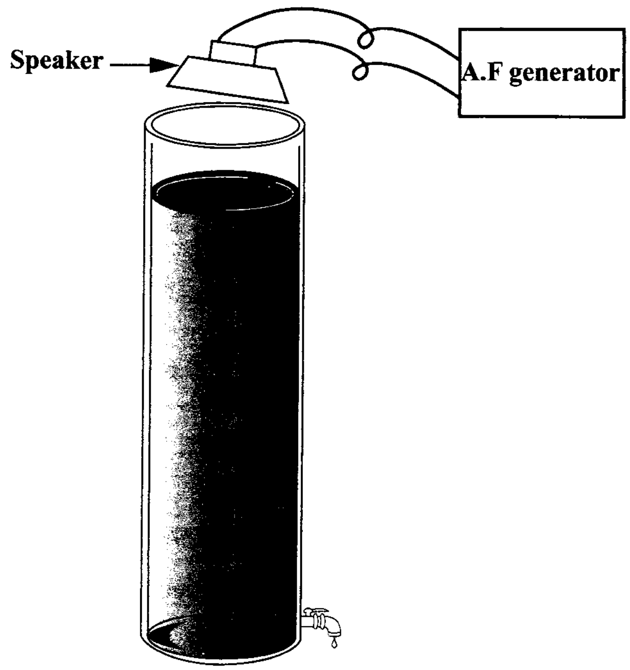

Figure 2

(i) Describe and explain what is heard at a certain level as the water runs out at the lower end.

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____

[2 marks]

(ii) What name is given to this phenomenon?

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[1 mark]

(b) The frequency of a vibrating string is given by $f = \frac{1}{2l}\sqrt{\frac{T}{\mu}}$, where T is the tension in the string, μ is the mass per unit length of the string and l is the length of the string.

While the tension of a vibrating string was kept constant at 100 N, its length was varied in order to tune the string to a series of tuning forks. The results obtained are shown in Table 2.

TABLE 2

Frequency of Fork, f (Hz)	256	288	320	384	450	512
Length of String, l (m)	0.781	0.695	0.625	0.521	0.444	0.391
$1/l$ (m^{-1})						

(i) Complete the table by filling in the values for $1/l$. **[1 mark]**

(ii) On page 9, plot a graph of frequency f vs $1/l$. **[3 marks]**

(iii) Use the graph to determine the frequency of an unmarked fork which was in tune with 41.7 cm of the string.

[2 marks]

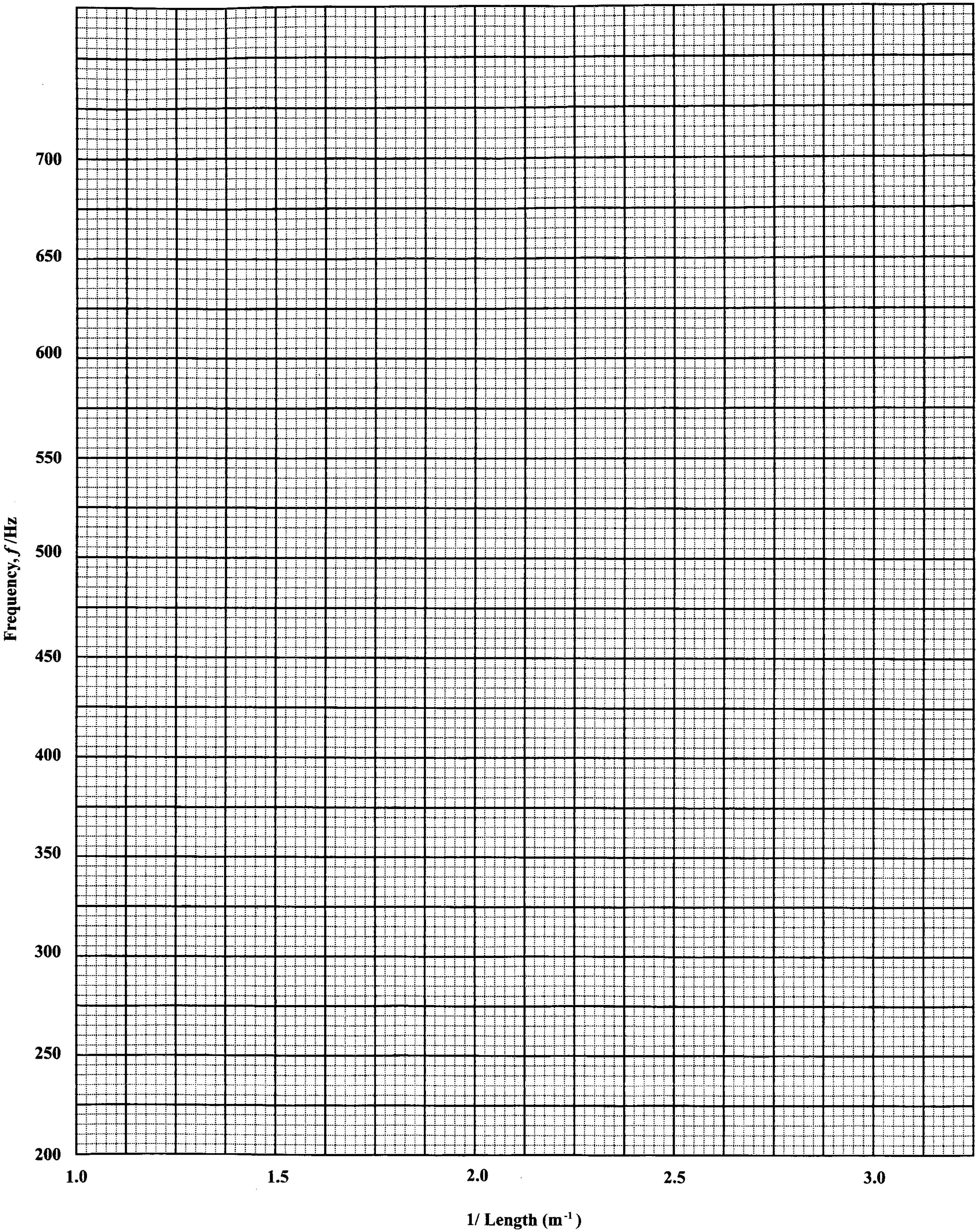
(iv) Calculate the gradient of the graph.

[3 marks]

(v) Determine the value of the mass per unit length, μ, of the wire.

[3 marks]

Total 15 marks


Frequency, f /Hz
700
650
600
550
500
450
400
350
300
250
200
1.0
1.5
2.0
2.5
3.0
1/ Length (m⁻¹)

3. (a) Define the term 'specific latent heat of fusion' of a substance.

____

____

[1 mark]

(b) A solid of mass 2.0 kg receives heat at the rate of 1.0×10^5 J per minute. Its temperature, T, (in K) during the first 45 minutes of the experiment is shown in the graph on the opposite page.

(i) What are the values of the melting point and boiling point of the substance?

Melting point ________________ Boiling point ____________________

[2 marks]

(ii) Calculate the gradient of the two linear regions of the graph labelled P and Q.

Gradient P

Gradient Q

[4 marks]

(iii) State, giving your reasoning, whether the specific heat capacity of the liquid is greater or smaller than that for the solid state.

____

____

[2 marks]

(iv) Calculate the specific heat capacity of the substance in the liquid state.

[3 marks]

(v) Calculate the specific latent heat of fusion of the substance.

[3 marks]

Total 15 marks

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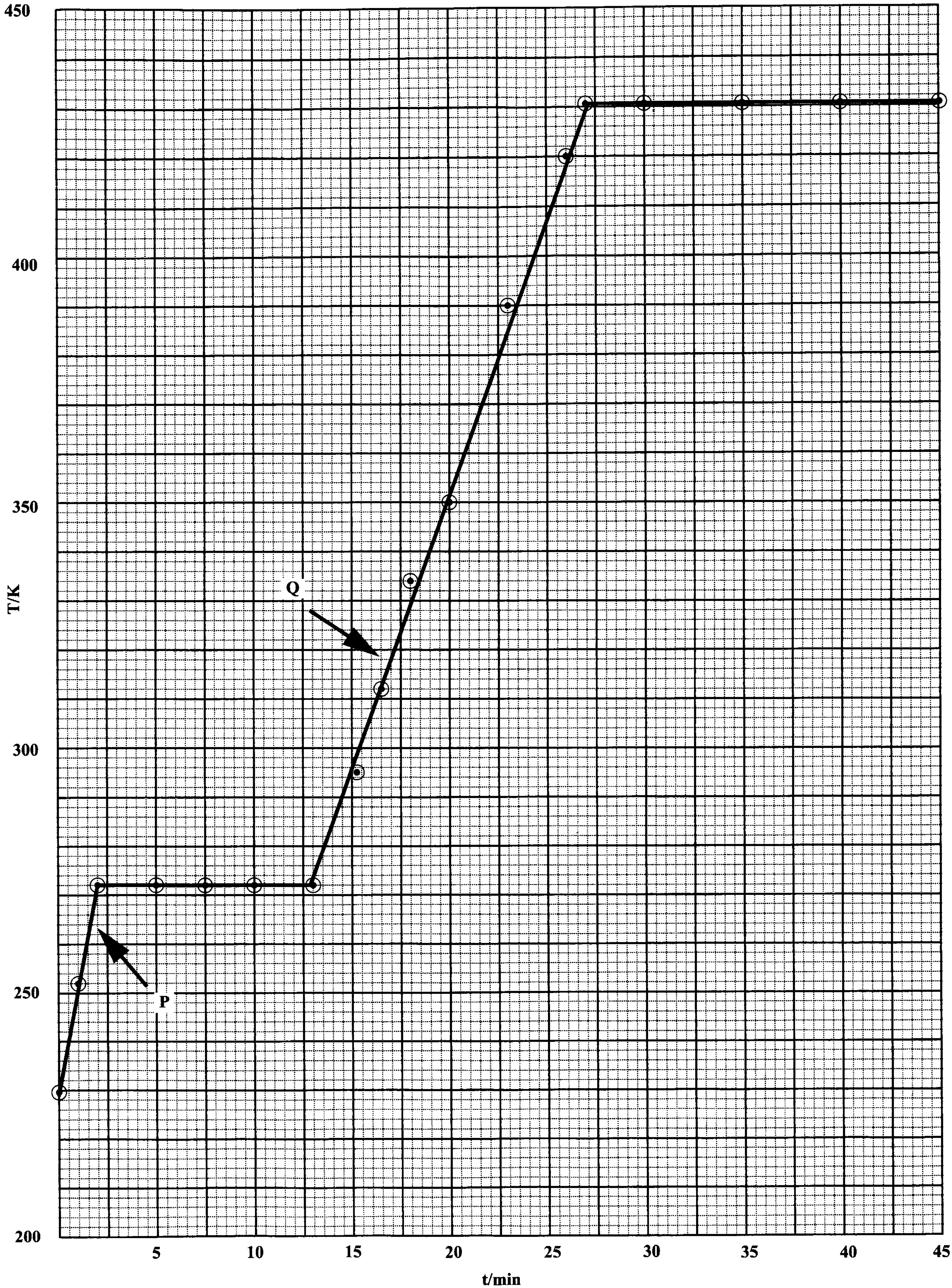
450
400
350
300
250
200
T/K
Q
P
5
10
15
20
25
30
35
40
45
t/min

SECTION B

Answer ALL questions.

Write your answers in the spaces provided at the end of the questions.

4. (a) (i) Distinguish between the 'kinetic energy' and the 'gravitational potential energy' of a body near the Earth's surface and write an expression for EACH.

(ii) Explain why it is possible for the TOTAL mechanical energy of a system to be negative, but NOT its kinetic energy.

[6 marks]

(b) Figure 3 shows a ski jump.

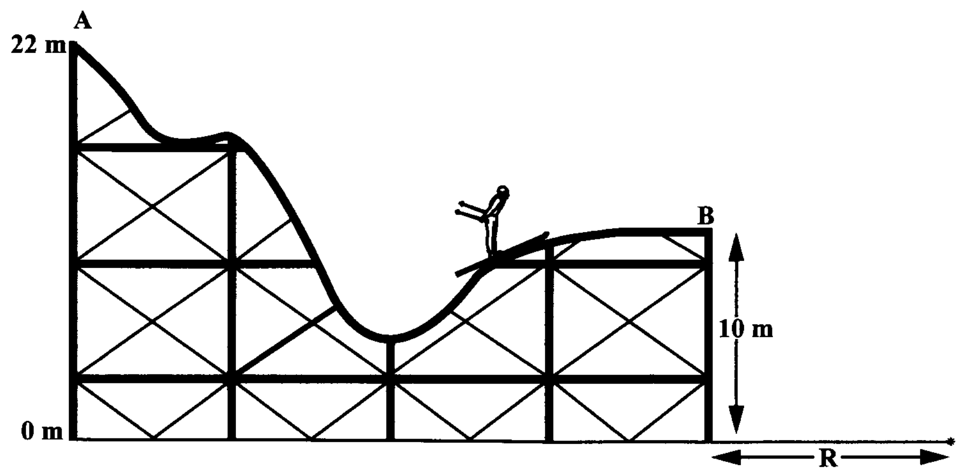

Figure 3

A ski jumper, of mass 60 kg, starts from rest at the top, A, of the jump track and follows, with its several ups and downs, the path of the track. The jump track starts at a height of 22 m above the eventual landing point. The jumper leaves the track while he moves horizontally at Point B which is exactly 12 m below the starting point.

Calculate the

(i) velocity of the skier at Point B, the end of the track

(ii) time the skier takes to hit the ground after he has left Point B

(iii) horizontal distance, R, of the end of the track, (Point B) from the landing point.

[7 marks]

(c) What TWO assumptions did you make in calculating the answers to (b) above?

[2 marks]

Total 15 marks

GO ON TO THE NEXT PAGE

You MUST write the answer to Question 4 here.

4. (a)

(b)

(c) ____

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____

5. (a) (i) Distinguish **clearly** between 'refraction' and 'diffraction' of light. Draw ray diagrams, ONE in EACH case, to illustrate EACH phenomenon. **[4 marks]**

(ii) Discuss how interference and diffraction contribute to the action of a diffraction grating. **[3 marks]**

(b) Light from glowing hydrogen contains two discrete spectral lines at 656.3 nm and 486.1nm. These two lines are viewed through a spectrometer using a diffraction grating with 6000 slits per cm. The spectra is viewed in the second order (n = 2).

Calculate the

(i) slit spacing of the diffraction grating, expressed in metres **[2 marks]**

(ii) angular positions of the two spectral lines **[5 marks]**

(iii) angular separation between the two spectral lines. **[1 mark]**

Total 15 marks

GO ON TO THE NEXT PAGE

You MUST write the answer to Question 5 here.

5. (a)

GO ON TO THE NEXT PAGE

(b)

6. Figure 4 shows some gas contained in a cylinder with a piston in thermal equilibrium at a temperature of 315 K. The cylinder is **thermally isolated from its surroundings**. Initially the volume of gas is 2.90×10^{-4} m^3 and its pressure is 1.03×10^5 Pa.

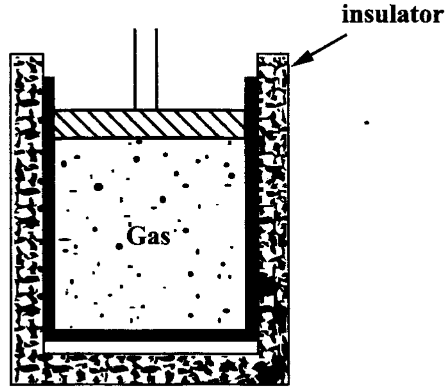

Figure 4

(a) Use the equation of state for an ideal gas to find the amount, in moles, of gas in the cylinder. **[3 marks]**

(b) The gas is then compressed to 3.5×10^{-5} m^3 and its temperature rises to 790 K. Explain this rise in temperature

(i) on a macroscopic scale, using the first law of thermodynamics **[3 marks]**

(ii) on a microscopic scale, using the kinetic theory of gases. **[3 marks]**

(c) Calculate the pressure of the gas after this compression. **[2 marks]**

(d) The work done on the gas during the compression is 90 J. Use the first law of thermodynamics to find the **increase** in the internal energy of the gas during the compression. **[1 mark]**

(e) Calculate the molar heat capacity of the gas in the container. **[3 marks]**

Total 15 marks

You MUST write the answer to Question 6 here.

6. (a)

(b)

(c)

(d)

(e)

END OF TEST